Eigenvalues of the Exact One-Step Matrix and the Eigen-Predicted Characteristic Time: 96×96, $\tau = 0.01$

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This note computes the two slowest decay rates λ_1 and λ_2 of the total mass directly from the solver's discretization, uses them to compute t^* without differencing the mass series, and compares that value with the right-sweep mask t^* already reported in `t_star_96x96_tau0.01.csv` in this folder. The four pairs whose $T = 4$ runs are still pending, $(10, 0.1)$, $(10, 1)$, $(100, 0.1)$ and $(100, 1)$, are not compared.

1 Setup

All quantities match job `0b8d4022` ($T = 1$) and job `54cdf1f3` ($T = 4$): a 96×96 polar grid ($r_g = r_y = 96$), $v = 1$, four tubes at `N_LIST` $= [0, 24, 48, 72]$, $d_{\text{tube}} = 0$, $D = 1$, unit radius, centred initial condition. The time step is $\Delta t = 0.1 \min(\Delta r^2, \Delta\theta^2 \Delta r^2)/(2D) = 2.3240 \times 10^{-8}$ (`compute_dT`), and the mass series is sampled every $5\Delta t$, so the sample spacing is $h = 1.1620 \times 10^{-7}$.

2 The exact one-step matrix

2.1 State vector

The solver's state after step k is collected into one vector $x_k \in \mathbb{R}^n$:

$$x = \big(c,\ \ \phi_{m,j}\ (0 \le m \le r_g - 2,\ 0 \le j < r_y),\ \ \rho_{m,j}\ (0 \le m < r_g,\ j \in \texttt{N_LIST}) \big),$$

where c is the central patch, ϕ the diffusive layer and ρ the advective layer on the four tube rays. The outermost diffusive ring $m = r_g - 1$ is the absorbing rim, held at zero, so it is not a state variable. This gives $n = 1 + (r_g - 1)r_y + 4r_g = 9{,}505$. The total mass is linear in the state, $M = w^{\mathsf{T}} x$, with the weights used by `calc_mass_diff` and `calc_mass_adv`: $\pi\Delta r^2$ for c, $(m + 1)\Delta r^2 \Delta\theta$ for $\phi_{m,j}$ and Δr for $\rho_{m,j}$. The initial state is $x_0 = (1/(\pi\Delta r^2), 0, \ldots, 0)$, so $M(0) = 1$.

2.2 One solver step is a linear map

One iteration of `advance_mass_analysis` (`analysis_tools.py:793`) calls `comp_DL_AL_kp1_2step` (`numerical_tools.py:23`), which writes the new ϕ and ρ from the old ones, then `u_center` (`numerical_tools.py:250`), which updates c from the old ϕ, ρ and c, and then copies the new layers over the old. Every update is a linear combination of the old state, with one exception: `u_center` adds $|v\rho_{0,j}|$ (`numerical_tools.py:278`). Physical states have $\rho \ge 0$, and the scheme keeps them nonnegative, so on every state the solver actually visits this term equals $|v|\rho_{0,j}$. The step is therefore exactly linear on physical states,

$$x_{k+1} = P\, x_k, \qquad P = I + \Delta t\, L,$$

where L is the semi-discrete operator. (The velocity is passed as $-v$, as both drivers do: `analysis_tools.py:869`, `gpu/driver.py:122`.)

2.3 Extracting P column by column

Column j of P is Pe_j, the result of one solver step applied to the j-th unit vector. For each $j = 1, \ldots, n$ the state is set to e_j, one step is taken with the repository's own `comp_DL_AL_kp1_2step` and `u_center`, unchanged, and the new state is stored as column j. No stencil is re-derived, so P is the discretization exactly as the code implements it. At 96×96 it has about five nonzeros per column, and $L = (P - I)/\Delta t$.

Checks.

- Linearity, at both 16×16 and 96×96: for random nonnegative x, y, $\|\mathrm{step}(x + 2y) - \mathrm{step}(x) - 2\,\mathrm{step}(y)\|_\infty \le 9 \times 10^{-16}$, and $\|Px - \mathrm{step}(x)\|_\infty \le 3 \times 10^{-16}$.
- Reproduction of stored runs: at 16×16, where the full $\ln M$ curves are in the repository (`2026-10-05_ab-grid-16x16_second-derivative/curves_16x16.csv` and `2026-10-05_ab-pairs-16x16_T4/curves_16x16_T4.csv`), iterating $x_{k+1} = Px_k$ from x_0 reproduces the stored $\ln M$ to within 6×10^{-12} for all sixteen (a, b) pairs.
- P is entrywise nonnegative. Apart from the four outermost tube cells, which receive no inflow and are never excited, its graph is strongly connected and contains x_0. By Perron–Frobenius the slowest mode is therefore real and simple.

3 Computing λ_1 and λ_2

3.1 Eigenpairs and modal amplitudes

Let $Lr_n = \mu_n r_n$ (right eigenvectors) and $z_n^\mathsf{T} L = \mu_n z_n^\mathsf{T}$ (left eigenvectors, obtained as eigenvectors of L^T). Expanding the initial state, $x_0 = \sum_n c_n r_n$ with $c_n = z_n^\mathsf{T} x_0 / z_n^\mathsf{T} r_n$, gives

$$M(t) = \sum_n a_n e^{-\lambda_n t}, \qquad a_n = (w^\mathsf{T} r_n)\, c_n.$$

The slowest modes are the eigenvalues of L nearest zero. They were computed with shift-invert Arnoldi (`scipy.sparse.linalg.eigs` with $\sigma = 0$, $k = 160$), once for L and once for L^T, with left and right eigenvectors paired by eigenvalue. Repeating the whole calculation with $k = 80$ changed t^* by at most 5.3×10^{-5}, so $k = 160$ is ample.

3.2 Decay rates that are exact for the discrete series

The stored series obeys $x_{k+1} = (I + \Delta t L)x_k$ exactly, so its modes evolve as ν_n^k with $\nu_n = 1 + \Delta t\, \mu_n$. The decay rates are therefore taken as

$$\lambda_n = -\frac{\log \nu_n}{\Delta t} = -\mu_n + O(\Delta t\, \mu_n^2),$$

which makes $\sum_n a_n e^{-\lambda_n t}$ agree with the time-stepped series at every step rather than only as $\Delta t \to 0$.

3.3 Only some modes are excited

The centred initial condition and the four-fold tube arrangement are symmetric, so most modes are not excited. At 96×96, for every pair compared here, the non-excited modes have amplitudes at round-off level, $|a_n|/|a_1| \le 2 \times 10^{-15}$, while the excited ones have $|a_n|/|a_1| \ge 2.8 \times 10^{-9}$; 38 of the 160 computed modes are excited. Modes with $|a_n| \le 10^{-9}|a_1|$ are discarded. The relevant λ_2 is therefore the slowest excited mode after λ_1, which is often not the second eigenvalue of L. For example, at 16×16 with $(a, b) = (0.1, 100)$, L's second eigenvalue, 15.21, has $|a_n| \sim 10^{-28}$; the excited λ_2 is 31.57. Table 1 lists the results. When λ_2 is complex, its conjugate is also a mode, and the pair together contributes $2|a_2|e^{-\mathrm{Re}\,\lambda_2 t}\cos(\mathrm{Im}\,\lambda_2\, t + \varphi)$ to M.

(a, b)	λ_1 (eigen)	λ_1 (tail fit)	rel. diff.	λ_2	R
(0.1, 0.1)	5.59230	5.59240	1.7×10^{-5}	$9.2324 \pm 4.0159\,i$	0.369
(0.1, 1)	5.67993	5.67992	1.4×10^{-6}	$10.0576 \pm 3.9223\,i$	0.324
(0.1, 10)	5.81972	5.81998	4.5×10^{-5}	$18.3273 \pm 3.4978\,i$	0.181
(0.1, 100)	5.82626	5.82625	1.7×10^{-6}	30.8115	0.660
(1, 0.1)	4.71291	4.71329	8.1×10^{-5}	$6.9734 \pm 4.8473\,i$	0.511
(1, 1)	4.98297	4.98280	3.3×10^{-5}	$7.7195 \pm 4.7561\,i$	0.459
(1, 10)	5.75924	5.75962	6.6×10^{-5}	$15.7730 \pm 4.0606\,i$	0.204
(1, 100)	5.82322	5.82321	1.7×10^{-6}	30.7936	0.659
(10, 10)	5.25509	5.25466	8.3×10^{-5}	$13.1578 \pm 5.0070\,i$	0.231
(10, 100)	5.79294	5.79293	1.8×10^{-6}	30.6140	0.656
(100, 10)	3.12987	3.12874	3.6×10^{-4}	$11.1202 \pm 5.4688\,i$	0.150
(100, 100)	5.50206	5.50204	2.4×10^{-6}	28.7945	0.615

Table 1: Slowest decay rate λ_1 and slowest excited subdominant rate λ_2 at 96×96. The tail fit is $-\ln 10 \times$`fit_slope_log10` from `t_star_96x96_tau0.01.csv`, a least-squares line through $\log_{10} M$ on $[t^*, T]$; it agrees with the eigenvalue to within 3.6×10^{-4}. $R = |a_2/a_1|$ for real λ_2 and $2|a_2/a_1|$ for a complex pair.

4 Computing t^* (eigen)

4.1 Exact derivatives of $\ln M$

With weights $p_n(t) = a_n e^{-\lambda_n t}/M(t)$, the derivatives of $y = \ln M$ follow directly from the modal sum:

$$y'(t) = -\sum_n p_n \lambda_n, \qquad y''(t) = \sum_n p_n \lambda_n^2 - \Big(\sum_n p_n \lambda_n\Big)^2.$$

So y'' is the weighted spread of the decay rates of the modes still present; it vanishes once a single mode remains. The weights are evaluated as $a_n e^{-(\lambda_n - \lambda_1)t}$, normalized, to avoid underflow at large t. No finite differences are involved.

4.2 The right-sweep rule on the exact y''

The rule is the committed one (`launch._char_time_onset`, `launch.py:1166`): t^* is the end of the last interval on which $|y''| > \tau$, and there is no t^* if $|y''(T)| > \tau$. It is applied with the same $\tau = 0.01$ and the same run length T as the measured value: $T = 4$ for the four finished $b \le 1$ pairs and $T = 1$ otherwise. In practice, $|y''|$ is evaluated on 50,001 points of $[0.01, T]$, the last point with $|y''| > \tau$ is found, and the root of $|y''(t)| - \tau$ just after it is refined by Brent's method to 10^{-12}.

4.3 Relation to λ_1 and λ_2

At late times only λ_1 and λ_2 matter. With $\delta = \lambda_2 - \lambda_1$,

$$y'' \approx R\,|\delta|^2 e^{-\operatorname{Re}\delta\, t} \times \begin{cases} 1 & \lambda_2 \text{ real}, \\ \cos(\operatorname{Im}\delta\, t + \varphi) & \lambda_2 \text{ complex}, \end{cases} \qquad \Longrightarrow \qquad t^* \approx \frac{\ln\big(R\,|\delta|^2/\tau\big)}{\operatorname{Re}\delta}.$$

For real λ_2 (the $b = 100$ pairs) this formula agrees with the exact t^* to within 2×10^{-6}. For complex λ_2 it is the time at which the envelope falls below τ; the mask's t^* falls up to half a period, $\pi/\operatorname{Im}\delta$, earlier. λ_1 itself enters t^* only through the gap δ; it sets the decay after t^*, $M(t) \approx a_1 e^{-\lambda_1 t}$.

5 Accuracy: t^* (eigen) against t^* (mask)

(a, b)	T	t^* (eigen)	t^* (mask)	Δ	$\lvert\Delta\rvert/h$	$\lvert y'''(t^*)\rvert$	$\lvert\Delta\, y'''(t^*)\rvert$
(0.1, 0.1)	4	1.8506990	1.8507035	-4.5×10^{-6}	38.7	0.063	2.8×10^{-7}
(0.1, 1)	4	1.5451757	1.5451862	-1.0×10^{-5}	90.3	0.023	2.4×10^{-7}
(0.1, 10)	1	0.4053995	0.4053999	-4.1×10^{-7}	3.5	0.386	1.6×10^{-7}
(0.1, 100)	1	0.4252720	0.4252720	-1.5×10^{-8}	0.1	0.250	3.8×10^{-9}
(1, 0.1)	4	2.8875764	2.8875777	-1.3×10^{-6}	10.9	0.113	1.4×10^{-7}
(1, 1)	4	2.3043555	2.3043566	-1.1×10^{-6}	9.6	0.141	1.6×10^{-7}
(1, 10)	1	0.7496629	0.7496647	-1.7×10^{-6}	15.0	0.070	1.2×10^{-7}
(1, 100)	1	0.4254582	0.4254584	-2.2×10^{-7}	1.9	0.250	5.4×10^{-8}
(10, 10)	1	0.8761470	0.8761477	-7.9×10^{-7}	6.8	0.155	1.2×10^{-7}
(10, 100)	1	0.4273314	0.4273317	-2.8×10^{-7}	2.4	0.248	7.1×10^{-8}
(100, 10)	1	0.7289982	0.7289984	-2.0×10^{-7}	1.7	0.234	4.6×10^{-8}
(100, 100)	1	0.4471435	0.4471439	-3.2×10^{-7}	2.8	0.233	7.6×10^{-8}

Table 2: t^* at 96×96, $\tau = 0.01$. t^* (mask) is the `t_star` column of `t_star_96x96_tau0.01.csv`: the right-sweep rule on the central second difference of the stored GPU series ($H = 861h \approx 1.0005 \times 10^{-4}$), reported at sample resolution. $\Delta = t^*(\text{eigen}) - t^*(\text{mask})$; $h = 1.1620 \times 10^{-7}$ is the sample spacing.

The two values agree to within 1.0×10^{-5} for all twelve pairs, at most 90 sample spacings. The mass at t^* agrees to within a relative 5.9×10^{-5}. Every Δ is negative: the mask's t^* is slightly later. Part of this is by construction: the mask reports the sample after the crossing (up to h later), and stored sample i holds the mass one solver step before $t = ih$ (a further $\Delta t = h/5$).

The size of Δ is governed by how steeply $|y''|$ crosses τ. A discrepancy $\delta y''$ in the second derivative moves the crossing by about $\delta y''/|y'''(t^*)|$. The last column, $|\Delta\, y'''(t^*)|$, is the discrepancy that would account for each Δ; it lies between 3.8×10^{-9} and 2.8×10^{-7} for every pair, within the 10^{-7} to 10^{-6} noise level of the second difference at $H \approx 10^{-4}$ stated in `launch.py`. The largest Δ occur where the crossing is shallowest: $(0.1, 1)$, where $|y'''(t^*)| = 0.023$ because t^* sits near the bottom of a trough of y'', and $(0.1, 0.1)$. Where the crossing is steep (the $b = 100$ pairs), Δ is a few sample spacings or less.

6 Reproducing

The probe, the eigensolve and the t^* computation are about 1.6 s per (a, b) pair at 96×96 on one CPU (probe 0.4 s, eigensolve with $k = 160$ 0.9 s, t^* 0.3 s), using the repository at commit `dca5c97` with Python 3.14, NumPy 2.5, SciPy 1.18 and Numba 0.68. The same procedure applies unchanged to other grids. All values here are for the 96×96 grid; convergence of λ_1, λ_2 and t^* under grid refinement is not addressed in this note.